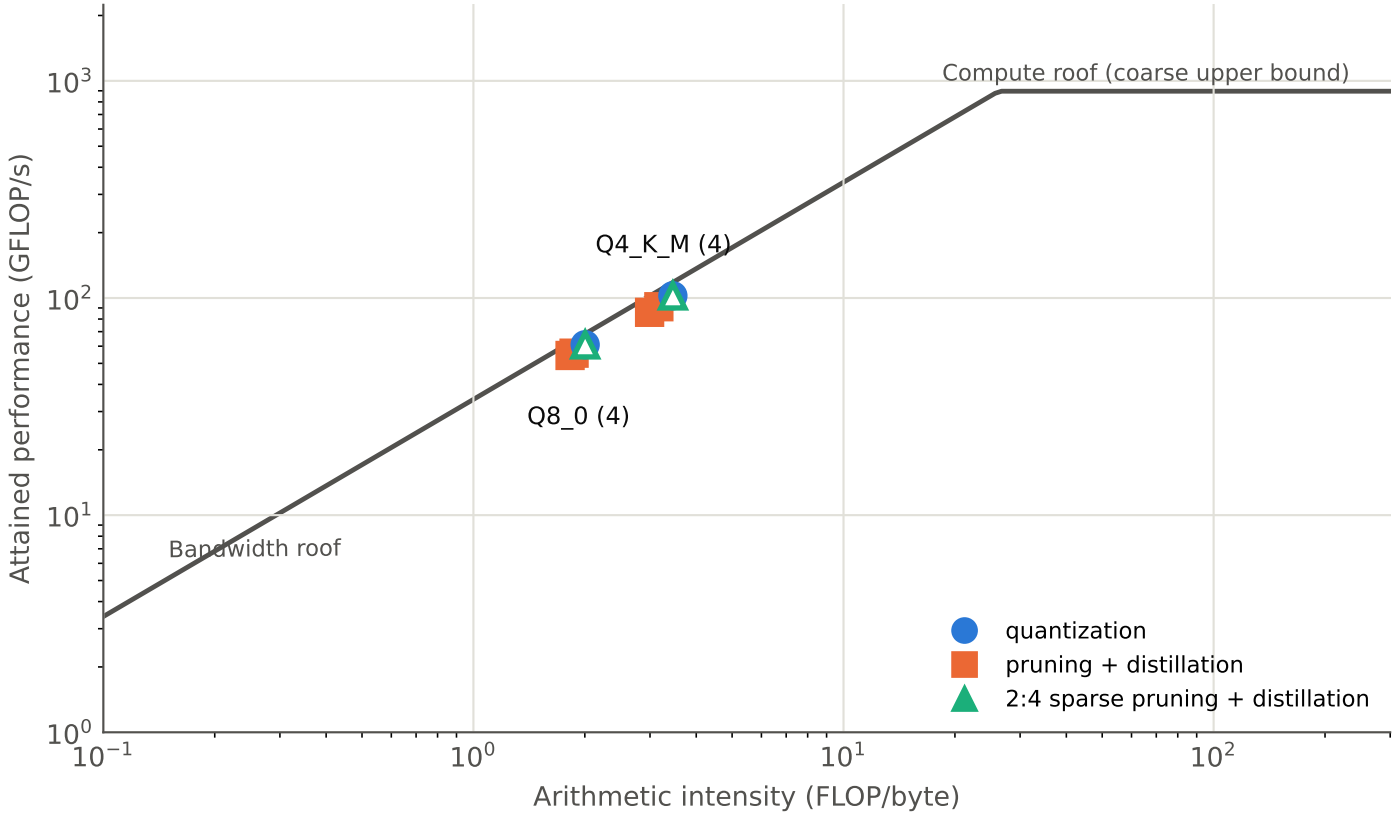

Coarse roofline: arithmetic intensity = 2 × parameters / corrected bytes per token; attained performance = 2 × parameters × tokens/s. The compute roof is the AVX2 fp32 FMA upper bound (8 cores × 3.5 GHz); quantized kernels do not reach it, so only the bandwidth roof is quantitatively meaningful.